

3 Types of Stretching and the Effects on Flexibility

There are a number of different types of stretching and the public can be confused regarding their differences. Passive and active static stretching, dynamic, ballistic, proprioceptive neuromuscular facilitation (PNF), and others are used to enhance flexibility or range of motion (ROM) as well as be incorporated as part of a pre-competition or training warm-up. First of all, what is the definition of flexibility? Michael J. Alter in his textbook (6) lists a variety of definitions. Some of the definitions of flexibility include:

- 1 ROM available to a joint or group of joints (50, 85, 93, 116, 185)
- 2 Total achievable excursion (within limits of pain) of a body part through its potential ROM (167)
- 3 Ability to move a joint smoothly and easily through its complete pain free ROM (100, 101)
- 4 Ability to move a joint through a normal ROM without undue stress to the musculotendinous unit (41)
- 5 Normal joint and soft tissue ROM in response to active or passive stretch (81)

The ability to increase a joint's ROM, would normally necessitate an improved extensibility (78) or decreased stiffness of musculotendinous and other connective tissues. Gajdosik and colleagues (65, 66) suggest that flexibility should be described as a ratio of change in muscle length or joint angle to a change in force or torque. Recent research use this description or technique by subjecting a limb joint to an extended ROM on an isokinetic dynamometer as a test for muscle stiffness (122, 124, 136, 139, 146).

To achieve these increases in flexibility, a variety of stretching techniques are used. Static stretching for example, involves lengthening a muscle until either a stretch sensation or the point of discomfort is reached and then holding the muscle in a lengthened position for a prescribed period of time (23, 24, 26, 48). Whether it is passive or active static stretching depends upon whether the muscle is lengthened by an external force (i.e., another person, or a tool like a stretching band or machine) with the muscle relaxed (passive static stretch) or lengthened by an active contraction of the affected muscle or other muscles (i.e., antagonist) (active static stretch) (see Figure 3.1). Static stretching is used in athletic, fitness,



Figure 3.1 Unassisted, and assisted passive static stretching with a partner or a band

health and rehabilitation environments. It is an effective method for increasing joint ROM (14, 15, 162) and was purported to improve performance (198, 199) and reduce the incidence of activity-related injuries (23, 24, 26, 126, 169, 170, 182) (primarily musculotendinous injuries associated with explosive and change of direction actions, 24, 29). However, the possibility of static stretch-induced performance impairments has limited its use in the new millennium. Evidence for and against this bias will be presented in a subsequent chapter.

Dynamic stretching uses a controlled movement through the ROM of the active joint(s) (61). It can be exemplified by swinging the legs back and forth (hip flexion and extension) or side to side (hip abduction and adduction) or swinging the arms in circles (shoulder circumduction) (see Figure 3.2). Dynamic stretching differs from ballistic stretching in that ballistic stretching would typically involve higher velocity movements with bouncing actions at the end of the ROM (10, 148). Ballistic movements were used in the aforementioned 5BX programme (popular from 1950s until the 1960s) and were still prevalent till quite recently in many military and police style training. In the late 1980s, I consulted with fitness instructors of the Royal Canadian Mounted Police at their national training depot in Regina, Saskatchewan, Canada. Recruits were trained and tested by performing resistance exercises such as shoulder presses, push-ups, sit-ups and others as quickly as possible in a prescribed time. The number of repetitions completed in 30 or 60 seconds



Figure 3.2 Dynamic stretching (hip extension and flexion)

was measured and thus the recruits would ballistically throw the barbells up and down (i.e., shoulder press) or slam their trunk back and forth (i.e., sit-ups) as quickly as possible. Definitely a recipe for injuries!

What is the difference between dynamic stretching and dynamic activity? It could be argued that dynamic stretching is a dynamic activity but not all dynamic activities are considered dynamic stretching. The decisive factor is whether the dynamic activity moves the body through a full or nearly full ROM. Jogging, skipping, hopping and other similar activities are all dynamic, but as they only emphasize a restricted or small to moderate ROM they would not be considered as dynamic stretching. However, if the person did butt (gluteal) kicks (knee flexion and touches the buttocks with heel of the foot) while jogging, then this dynamic activity would be under the purview of dynamic stretching as it goes through a fuller ROM. As mentioned in the historical section, dynamic stretching in this millennium was considered preferable to static stretching in a warm-up, owing to training specificity (training movement matches the sport or exercise movements) (28, 172) as well as activity-induced increases in metabolism, muscle temperature (30, 62, 197, 198), and neural activation (73, 74, 75). Many systematic warm-up routines such as the FIFA 11 (football/soccer specific warm-up) utilize a combination of dynamic stretching and activities in preparation for subsequent activity and with evidence for decreased injury incidence (1, 2, 82, 149, 155, 180, 181).

PNF stretching combines static stretching and isometric contractions in a cyclical pattern. PNF was developed in the late 1940s and early 1950s by Herman Kabat, and two physical therapists, Margaret Knott and Dorothy Voss. Kabat, was a neurophysiologist, who developed PNF based on the neuromuscular research of Sir Charles Sherrington (178, 179). Two of the more ubiquitous techniques are the contract relax (CR) and contract-relax-agonist-contract (CRAC) techniques (168,



Figure 3.3 Proprioceptive Neuromuscular Facilitation (PNF) stretching for the hamstrings

177). The CR method includes a static stretching component followed immediately by an isometric contraction of the stretched muscle, followed with another stretch of the target muscle. CRAC involves an additional contraction of the agonist muscle (i.e., opposing the muscle group being stretched) during the stretch, prior to the additional stretching of the target muscle (see Figure 3.3). PNF was used extensively by team athletes (PNF stretches typically need a partner) in the late 20th century, but similar to static stretching, its use has diminished in this millennium. A number of individual studies suggest that PNF is more effective than static or dynamic stretching for improving ROM (70, 87, 154). However, a recent meta-analysis systematic review reported an advantage for static stretch training over PNF training for increasing ROM (187).

Why should we stretch? Stretching is primarily used to increase joint ROM. When used as part of a warm-up, the increased ROM was thought to improve performance and decrease the incidence of injuries (24, 26, 197). Although, the ability to increase ROM through stretching is generally universally agreed, its impact on performance and injuries is more controversial. There are many factors to consider including the type, duration (volume), intensity of stretching, the population that is stretching (i.e., male, female, young, old, athletic, sedentary and others) among other factors. Others may use stretching to achieve a greater sense of relaxation (i.e., yoga) (psychological or neurological effects).

Stretching-Induced Changes in Range of Motion

First of all, flexibility is not a global phenomenon of the body (7) but stretching effects can be global (20, 22). That is, an individual can be flexible in one joint

but not others (7). Even within a joint they may have a greater ROM with one movement versus another. A baseball pitcher may not exhibit similar relative flexibility as a discus thrower when comparing shoulder horizontal extension versus shoulder internal and external rotation. Some joints are more susceptible to stretch-induced increases in ROM. For example, stretching calf muscles provides only small increases in ankle dorsiflexion, which may not be clinically meaningful (164). In a subsequent chapter, we will discuss the global effects of stretching; that is: unilaterally stretch one muscle or joint and there can be increased ROM in another non-stretched muscle or joint.

What factors restrict our joint flexibility? Muscle fascicles (structural proteins like myosin, actin, titin and others), tendons, aponeuroses, joint capsules, and ligaments contribute to ROM restrictions with passive muscle elongation. As the composition of a human can range from 50–78 percent fluid (water) dependent on your age, sex, hydration, and other factors, the viscosity of the tissues can substantially affect flexibility. Furthermore, a highly activated central nervous system could increase muscle tonus and with a less relaxed muscle inhibit flexibility. In addition, pushing a joint to its maximum ROM can be uncomfortable or painful and thus the ability to tolerate this discomfort or pain may allow some individuals to stretch farther than other more pain sensitive individuals (121).

Stretching can induce elastic or plastic changes of the musculotendinous system. Elastic changes are defined as the elongation of tissues that recovers when the tension is removed; so, it is a temporary increase in ROM. Plastic changes involve a musculotendinous elongation where the tissue deformation remains even after the tension is removed; so, it is considered a semi-permanent change in flexibility. Elastic changes are the increases in ROM that are experienced immediately after a single session of stretching. A single session of static stretching can induce small-to-large relative ROM increases that persist for five (193), ten (23, 25, 27), 30 (63), 90 (102), and 120 minutes (162). Evidence of persistent improved ROM even one day later has been reported (137). However, not all studies show prolonged ROM increases. Two 15-second passive stretches did not provide a significant improvement in ROM at any time post-stretch (200), whereas augmented ROM was only evident for 3 minutes following four 30-second static stretches (52) and only for six minutes following five-modified hold and relax stretches (184). Specific training-induced plastic changes and mechanisms will be discussed in a later section.

As mentioned previously, there is much conflict as to what type of stretching provides the greatest ROM. PNF has been reported to provide greater acute or elastic ROM improvements than static stretching in some studies (57, 60, 151), which contrasts with other studies finding similar ROM changes between PNF and static stretching (46, 120). Dynamic stretching is also controversial with some studies reporting that a single session of dynamic stretching provides either similar (19, 158) or greater (8, 56) ROM increases as static stretching. But once again, conflicting studies report that dynamic stretching is not as effective as static stretching when used as part of warm-up

(10, 16, 156, 174, 176). When comparing static to ballistic stretching in a single (acute) stretching session, ballistic stretching has been reported to provide less ROM than static (10), or PNF (16), but similar ROM as static stretching and PNF in another study (104). Typically, we rely on systematic meta-analytical reviews to clarify the confusing messages from the wide variety of findings from small sample original research studies. A recent meta-analysis (21) of 47 studies revealed that a single session of stretching on average provided a small magnitude increase in ROM compared to non-active (non-stretched) control conditions. Whereas there were ROM increases with sit and reach, hamstrings, and triceps surae tests, there was no change with the hip adductor flexibility. It was also noted that stretch intensity, trained state, stretching techniques, or sex did not significantly modulate the enhanced ROM. A meta-regression demonstrated no relationship between the ROM increases with age or stretch duration.

In contrast to acute responses, is there a different response when examining chronic training study results? Another recent meta-analysis by our group (103) of 77 stretch training studies reported that chronic stretch training effects induced moderate magnitude ROM increases. With stretch training (in contrast to acute stretching effects) there were significant differences between the stretching techniques, with PNF and static stretching exhibiting greater increases in ROM than ballistic/dynamic stretching. Furthermore, females showed greater ROM training gains compared to males, whereas stretch volume, intensity, or frequency did not play a significant role in ROM gains. A 2018 meta-analysis systematic review determined only trivial differences in ROM improvements between acute and chronic PNF and static stretching (127), whereas another 2018 review found that chronic static stretch training provided greater flexibility than PNF training (187). Ballistic stretching (higher velocity movements with bouncing actions at the end of the ROM), which can be considered a variation of dynamic stretching was not as effective as static stretching for improving ROM after a 4-week training programme (47). Only one study compared PNF and dynamic stretching and indicated that after 14 training sessions that PNF provided 3–7 percent greater ROM increases (192).

Another form of stretching is termed neuromobilization. With neuromobilization, the nerve is placed under a lengthening stress. For example, when performing a seated straight leg raise, the individual would actively dorsiflex the foot and flex the cervical spine to exert a type of neural traction (Figure 3.4). In an 8 week stretch training study, neuromobilization provided a greater increase in hamstring length versus PNF in the first 4 weeks but in the last 4 weeks, hamstrings length actually decreased. Thus, overall, passive static stretching provided the greatest benefits (58).

Our psychological state or emotions can affect flexibility. Two sessions of 20 minutes each of anti-anxiety techniques such as the neuro-emotional technique, (also known as a mind body technique) was shown in one study to enhance ROM to a greater extent than a similar duration of passive static stretching or no stretching (96). It is purported that latent anxiety, which even



Figure 3.4 Neuromobilization

if not presently apparent can lead to learned emotional responses that affect the motor responses. Anti-anxiety techniques would dissipate these responses and lead to greater muscle relaxation and lower muscle tonus. Various forms of yoga combine the physical aspects of stretching with meditation or relaxation techniques to enhance the effects on flexibility.

Yoga

Serious yoga practitioners are widely known for their enhanced flexibility (161). However, yoga is also reported to improve a wide array of physiological and psychological parameters. For instance, it has been reported to increase muscular strength (119, 189) and endurance (189), balance (161), maximal oxygen uptake (189), improved reaction times (119), breath holding times (119), reduced cardiovascular risk, blood pressure, body mass index (13), unify the body, mind, and spirit, among others. How can yoga stretching accomplish all these beneficial measures? Yoga misperception is the problem. Yoga is not just stretching but involves many other components. In addition to the stretching aspect, full yoga practitioners should experience changes in their mental attitude (i.e., meditation), diet, practice of specific techniques for postures (*asanas*) and breathing practices (*pranayamas*) in order to attain a higher level of consciousness. It is unlikely that the average North American or European who goes to a trendy hot yoga session or most yoga sessions is serious or committed to a degree that they will attain a new level of consciousness. However, 30–60 minutes of stretching, changing positions and holding various postures (dynamic and static muscle endurance) and inspiratory and expiratory breathing techniques will certainly enhance a variety of

physiological and health parameters. From personal experience, having only participating in a single 60-minute yoga session, I experienced exercise-induced muscle soreness and delayed onset muscle soreness for days thereafter. Although I am considered a relatively very fit individual for my age, I was not accustomed to the prolonged and slow eccentric contractions when moving from one posture to the next or the ability to hold certain postures under extended muscle positions for prolonged periods. A lack of task specificity was certainly evident in my muscle pain over the next few days. Hence, you should not directly compare yoga to stretching practices like static dynamic, ballistic and PNF, as stretching is only one component of yoga. Furthermore, there are many types of yoga that may place greater emphasis on holding positions for longer durations to achieve changes in connective tissue (e.g., yin yoga) or place greater emphasis on breathing patterns or meditation.

Unfortunately, science is messy and results are not always consistent. The same interventions and measures can be used on sedentary women and men, youth, young adult and seniors, trained and untrained and you can get different results in every study. How do you figure out what is right? How do you know what to do? As mentioned, meta-analyses are usually considered the gold standard for integrating all the disparate information in the literature into a cogent understandable main message. But the problem with meta-analyses is that sometimes, they might hide some intricacies. Maybe, for the general population of sedentary and recreationally trained people, static and PNF stretching provide the greatest ROM increases. But as gymnasts and figure skaters and circus acrobats are the extremes of the population maybe just for them ballistic stretching is more important, as it emulates the actions in their sport. I just used those extreme athletes as a fictional example. There is very little research on such a small population of extreme flexibility athletes. If a meta-analysis looked at 30, 50, or 100 papers and only 2–3 papers dealt with such highly trained athletes then their responses to specific stretching could be hidden within the greater numbers of the other studies. A good meta-analysis should highlight these outliers but not all reviews accomplish it.

Thus, based on the latest information, it would seem that there is very little difference between stretching techniques when examining a single stretching session but with repetitive stretch training, PNF and static stretching might provide greater increases in ROM than dynamic stretching. However, we will examine later how prolonged static and PNF stretching without a full warm-up protocol might impair subsequent performance. Thus, you need to ask yourself, whether you need the utmost ROM. If you are jogging, the amplitude of your stride length is limited and there is no need to have an extreme ROM. Hence, some dynamic stretching before the running might be sufficient. If you are resistance training, and for your squat warm-up you take a light weight or just your body mass and go through a full ROM for 10 repetitions then that might be sufficient as you will not exceed that range while lifting. A study by Morton et al. (141) compared five weeks of resistance

training to static stretch training and found improved ROM in both groups with no significant differences between the groups in the flexibility of the hamstrings, hip flexors, extensors or should extensors. Thus, the ROM with resistance training was sufficient to provide a similar flexibility training adaptation as a stretching programme. We will delve into this topic in more detail later as one of our recent meta-analyses has shown that resistance training can provide similar increases in ROM as static stretching (3). Another study incorporated dancers who either resistance trained, stretched at a low intensity (3/10) or a moderate to high intensity (8/10) for six weeks (195). All groups improved their passive ROM with no difference between groups, whereas the resistance trained and low intensity stretch training groups improved their active ROM. The authors suggested that dance instructors and coaches should incorporate stretching and end-of-range resistance training within their schedules. Stretching would be incorporated at the end of the recovery session. They also recommended that the position of the stretches is very important in order to eliminate muscle contractions and hence the body should be in a stable position without extraneous tension. Each stretch according to this study should be held for 60 seconds at an intensity of 3/10.

The warm-up is not a time to try to make plastic (semi-permanent) changes in your flexibility. The warm-up prepares you for the upcoming activity at hand. There is no need to be able to do the Russian splits (legs completely abducted until the legs are horizontal on the floor) before a 5-km jog or step-ups in the weight room. However, if you have back problems because you spend eight hours per day sitting at a computer and now your pelvis has an anterior tilt, owing to shortened hip flexors affecting your lumbar spinal curvature, then the use of static or PNF stretching as a separate flexibility workout might be in order.

Range of Motion Norms

The average passive joint ROM have been provided in a few studies. The following tables provides a comparison of a sample of these studies in healthy or normal individuals.

Measuring Range of Motion

There are a myriad of instruments that can be used to measure ROM. Most individuals do not have access to advanced scientific laboratories. Hence, the universal, full circle goniometer is one of the most preferred pieces of equipment for measuring ROM (65) (see Figure 3.5). The use of goniometers has been evaluated as having high reliability (intraclass correlation coefficients of >0.91 (65). While instruments can have strong time to time (intra-rater) reliability they may not have good between instrument reliability. For instance, a universal, fluid and electronic goniometer were tested for reliability (71).

Table 3.1 Upper Limb and Back Passive ROM in degrees

<i>Joint</i>	<i>Motion</i>	<i>Family Medical Practice</i>	<i>Heyward 2005 (86)</i>
Shoulder	Flexion	180	150–180
	Extension	45–60	50–60
	Abduction	150	180
	External Rotation	90	90
	Internal Rotation	70–90	70–90
Elbow	Flexion		140–150
	Extension		0
Radioulnar	Pronation		80
	Supination		80
Wrist	Flexion		60–80
	Extension		60–70
	Radial Deviation		20
	Ulnar Deviation		30
Cervical spine	Flexion		45–60
	Extension		45–75
	Lateral Flexion		45
	Rotation		60–80
Thoraco-Lumbar Spine	Flexion		60–80
	Extension		20–30
	Lateral Flexion		25–35
	Rotation		30–45

Source: Family Medical Practice: (www.fpnotebook.com/Ortho/Exam/ShldrRngOfMtn.htm).

While the inter-tester reliability in Goodwin's study was excellent ($r = 0.90\text{--}0.93$), the reliability scores between instruments were not as consistent.

Fluid Goniometer vs. Universal Goniometer 0.90

Fluid Goniometer vs. Electrogoniometer 0.33

Universal Goniometer vs. Electrogoniometer 0.51

Another study compared a standard plastic goniometer to a fleximeter (gravitation-based ROM measuring device) or inclinometer (see Figure 3.6). The fleximeter/inclinometer demonstrated moderate to excellent intra- and inter-rater reliability but the goniometer showed poor to moderate intra- and inter-rater reliability (43). These findings indicate that if just one device is used to

Table 3.2 Lower Body Passive ROM in degrees

<i>Joint</i>	<i>Motion</i>	<i>Roass and Andersson 1982 (165)</i>	<i>AAOS 1969 (186)</i>	<i>Boone and Azen 1979 (33)</i>	<i>Heyward 2005 (86)</i>	<i>Hallaceli et al. 2014 (80)</i>
Hip	Extension	9.5	28	12.1	30	19.8
	Flexion	120.4	113	121.3	100–120	128.8
	Abduction	38.8	48	40.5	40–45	45.7
	Adduction	30.5	31	25.6	20–30	24.2
	Internal Rotation	32.6	35	44.4	40–45	43.4
	External Rotation	33.7	48	44.2	45–50	41.9
Knee	Extension		10		0–10	7.53
	Flexion	143.8	134	141	135–150	142.4
Ankle	Extension (Dorsi- flexion)	15.3	18	12.2	20	22.5
	Flexion (Plantar flexion)	39.7	48	54.3	40–50	49.99
	Valgus (Eversion)	27.9	18	19.2	15–20	19.9
	Varus (Inversion)	27.8	33	36.2	30–35	34.1

Source: American Association of Orthopaedic Surgeons.

measure differences before and after stretching or training, the extent of change should be reliable but you cannot always interchangeably use these devices to monitor flexibility differences. Similar conclusions were made when comparing goniometers and a digital level (143). Whereas, intra-tester reliability ranged from 0.91–0.99, inter-tester reliability ranged from 0.31 to 0.95 with limits 2.3 times higher for inter-tester reliability when testing for various shoulder movements (external and internal rotation and flexion). The authors indicated that experienced individuals using the same instrument for repeated measures (goniometer or digital level) should be able to detect a shoulder ROM change of at least 6 degrees, but when comparing measures from two people, the detectable change is 15 degrees. Another study (84) examined the reliability of measuring ROM with visual estimation, goniometry, still photography, “stand and reach” and hand behind back reach for six different shoulder movements. In general, they reported fair to good reliability ($r = 0.53$ – 0.73) for visual estimation, goniometry, still photography and stand and reach. However, the tests had standard errors of measurement between 14–25 degrees (inter-rater trial) and 11–23 degrees (intra-rater trial). The hand

Table 3.3 Norms for Joint Range of Motion (°)

MALES				FEMALES						
Joint Movement	Low	Mod. Low	Avg.	Mod. High	High	Low	Mod. Low	Avg.	Mod. High	High
Neck										
Flexion/ extension	<107	107–128	129–142	143–160	>160	<125	125–141	142–160	161–177	>177
Lateral Flexion	<74	74–89	90–106	107–122	>122	<84	84–99	100–116	117–132	>132
Rotation	<141	141–160	161–181	182–210	>201	<158	158–177	178–198	199–218	>218
Shoulder										
Flexion/ Extension	<207	207–223	224–242	243–259	>259	<226	226–242	243–261	262–278	>278
Adduction/ Abduction	<158	158–171	172–186	187–200	>200	<167	167–180	181–195	196–209	>209
Rotation	<154	154–171	172–192	193–210	>210	<289	189–206	207–227	228–245	>245
Elbow										
Flexion/ Extension	<133	133–143	144–156	157–167	>167	<133	133–143	144–156	157–167	>167
Forearm										
Supination/ Pronation	<151	151–170	171–191	192–211	>211	<160	160–179	180–200	201–220	>220
Wrist										
Flexion/ Extension	<112	112–131	132–152	153–172	>172	<136	136–155	156–176	177–196	>196
Ulnar/radial deviation	<64	64–77	78–92	93–105	>105	<75	75–88	89–101	102–117	>117

Table 3.3 (Cont.)

MALES		FEMALES								
Joint Movement	Low	Mod. Low	Avg.	Mod. High	High	Low	Mod. Low	Avg.	Mod. High	High
Hip										
Flexion/Extension	<50	50-67	68-88	89-106	>106	<82	82-99	100-120	121-138	>138
Adduction/Abduction	<41	41-50	51-61	62-71	>71	<45	45-54	55-65	66-75	>75
Rotation	<59	59-78	79-99	100-119	>119	<90	90-109	110-130	131-150	>150
Knee										
Flexion/Extension	<122	122-133	134-146	147-157	>157	<134	134-144	145-157	158-168	>168
Ankle										
Plantar flexion/dorsiflexion	<48	48-58	59-71	72-82	>82	<56	56-66	67-79	80-90	>90
Inversion/Eversion	<30	30-41	42-56	57-68	>68	<39	39-50	51-65	66-77	>77
Trunk										
Flexion/Extension	<45	45-62	63-83	84-101	>101	<30	30-47	48-68	69-86	>86
Lateral Flexion	<75	74-89	90-106	107-122	>122	<104	104-199	120-136	137-152	>152
Rotation	<108	108-126	127-147	148-166	>166	<134	134-152	153-173	174-192	>192

Source: Using Leighton Flexometer (88, 114).

Table 3.4 Percentile ranks for the Sit-and-Reach Test (cm) (88)

% Rank	AGE (Years)									
	20-29		30-39		40-49		50-59		60-69	
	M	F	M	F	M	F	M	F	M	F
90	39	40	37	39	34	37	35	37	32	34
80	35	37	34	36	31	33	29	34	27	31
70	33	35	31	34	27	32	26	32	23	28
60	30	33	29	32	25	30	24	29	21	27
50	28	31	26	30	22	28	22	27	19	25
40	26	29	24	28	20	26	19	26	15	23
30	23	26	21	25	17	23	15	23	13	21
20	20	23	18	22	13	21	12	20	11	20
10	15	19	14	18	9	16	9	16	8	15

Table 3.5 Percentile Ranks for the Modified Sit-and-Reach Test (88)

% Rank	FEMALES							
	<18 Years		19-35 Years		36-49 Years		>50 Years	
	Inches	cm	Inches	cm	Inches	cm	Inches	cm
99	22.6	57.4	21.0	53.3	19.8	50.3	17.2	43.7
95	19.5	49.5	19.3	49.0	19.2	48.8	15.7	39.9
90	18.7	47.5	17.9	45.5	17.4	44.2	15.0	38.1
80	17.8	45.2	16.7	42.4	16.2	41.1	14.2	36.1
70	16.5	41.9	16.2	41.1	15.2	38.6	13.6	34.5
60	16.0	40.6	15.8	40.1	14.5	36.8	12.3	31.2
50	15.2	38.6	14.8	37.6	13.5	34.3	11.1	28.2
40	14.5	36.8	14.5	36.8	12.8	32.5	10.1	25.7
30	13.7	34.8	13.7	34.8	12.2	31.0	9.2	23.4
20	12.6	32.0	12.6	32.0	11.0	27.9	8.3	21.2
10	11.4	29.0	10.1	25.7	9.7	24.6	7.5	19.1

% Rank	MALES							
	<18 Years		19-35 Years		36-49 Years		>50 Years	
	Inches	cm	Inches	cm	Inches	cm	Inches	cm
99	20.1	51.1	24.7	62.7	18.9	48.0	16.2	41.1
95	19.6	49.8	18.9	48.0	18.2	46.2	15.8	40.1
90	18.2	46.2	17.2	43.7	16.1	40.9	15.0	38.1
80	17.8	45.2	17.0	43.2	14.6	37.1	13.3	33.8
70	16.0	40.6	15.8	40.1	13.9	35.3	12.3	31.2

Table 3.5 (Cont.)

60	15.2	38.6	15.0	38.1	13.4	34.0	11.5	29.2
50	14.5	36.8	14.4	36.6	12.6	32.0	10.2	25.9
40	14.0	35.6	13.5	34.3	11.6	29.5	9.7	24.6
30	13.4	34.0	13.0	33.0	10.8	27.4	9.3	23.6
20	11.8	30.0	11.6	29.5	9.9	25.1	8.8	22.4
10	9.5	24.1	9.2	23.4	8.3	21.1	7.8	19.8

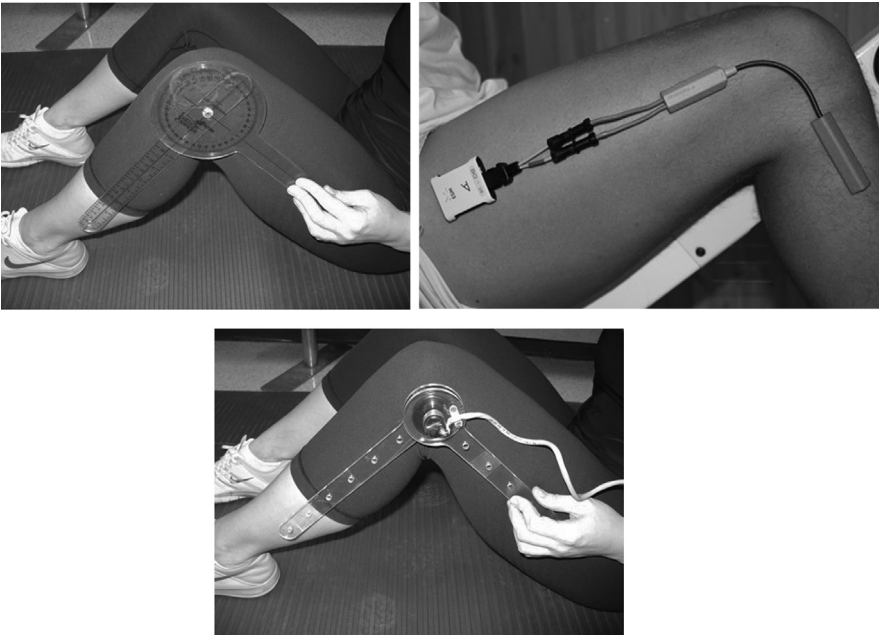


Figure 3.5 Standard and Electronic Goniometers

behind the back test showed poor inter-rater and intra-rater reliability ($r=0.14-0.39$). These poor findings are probably related to the number of joints involved and the movement complexity. Not all studies recommend visual inspection as van de Pol (190) stated that measurements of physiological ROM using goniometers or inclinometers were more reliable than using vision. However, another review reported that inter-rater reliability of lower extremity passive ROM measurement is generally low (191). One of the major problems is the sensation or measurements of end-feel. End feel is a term used to describe the extent of sensation on the hands of the examiner when they move the passive joint to the supposed end of ROM. Active ROM is reported to have higher reliability than passive ROM, as passive measures depend on the force applied to the limbs by



Figure 3.6 Inclination meters

another individual and that force and sensations can change unconsciously between trials and significantly between individuals.

Sometimes ROM tests may not measure what most people think they should measure. The ubiquitous sit and reach test is commonly used to measure lower back and hamstrings flexibility (see Figure 3.7). While a number of studies report that the sit and reach has moderate (95, 125) or strong (94) validity for measuring hamstrings extensibility but low validity for lumbar spine extensibility (117), another study contradicts and states that sit and reach (as well as the toe touch test) is a more appropriate measure of lumbar spinal flexibility and pelvic tilt ROM but not appropriate for hamstrings flexibility (144). These results instil confusion among the general population. The sit and reach test, and toe touch test can conceivably place undue pressure upon the vertebral discs (39) leading to injury. These concerns led to the development of the “back saver” sit and reach test, where only one leg (hamstring) is stretched at a time (157). Another variation is the chair sit and reach test (reliability: $r = 0.76\text{--}0.81$), where the individual while seated on the



Figure 3.7 Sit and reach test

front edge of the chair extends one leg with other leg flexed to the side (97). The individual similarly reaches as far forward as possible. Furthermore, scapular abduction or arm length differences could also affect sit and reach scores (90). Hence, a “modified” sit and reach (91) was introduced where the score is based on the difference between the initial starting touch point of the fingers on the device from a straight back (good posture) seated position and their maximum reach rather than just the maximum reach, which is commonly measured in the non-modified sit and reach tests. Finally, it has also been argued that taller individuals can reach farther than shorter individuals with the sit and reach test, as a longer spine permits greater spinal flexion (67). Obviously, no test is perfectly valid or reliable but, in general, the articles tend to indicate that if you use one particular test throughout, then it will be fairly reliable at detecting change over time, but it is difficult and precarious to compare the scores between one test and another.

It is very important when monitoring flexibility to ensure that the joint or joints to be measured are isolated and that other joints are not contributing to a functional ROM. Individuals who lack flexibility or ROM in one or a series of joints sometimes can compensate by using or emphasizing another part of their body that has better flexibility. This compensatory relative flexibility (171) assumes that the individual will want to move through a movement or ROM with the least resistance. Thus, when measuring hip extensor ROM in a pronated position, if the individual has tight quadriceps, they may extend their lumbar spine to achieve a greater score or if measuring supine hip flexor ROM, with tight hamstrings they could posteriorly tilt their pelvis to help increase the movement of their leg. When performing an activity, the individual may compensate for their lack of flexibility by relying on another body segment not well designed for the movement and could lead to injuries (6).

Sex Differences

It is common knowledge that most women have better flexibility than the average man (4, 64, 79, 98, 183, 196). Some factors contributing to this difference could be differences in muscle mass, joint geometry and the degree of collagen in the musculotendinous unit (6). For instance, men with highly hypertrophied biceps brachii or hamstrings may be restricted by the muscle mass from achieving a full elbow flexion or knee flexion ROM. Adolescent female volleyball players demonstrated greater shoulder internal rotation ROM than men but there was no association of shoulder injuries with more restricted shoulder ROM (135). Women show greater hip flexibility with a single leg raise test (131) and sit and reach tests (115). The broader and shallower hips of women contribute to this greater flexibility (6). Pelvic and thoracic angles are also greater in women (118). Not all tests ratify these assumptions. For example, in one study, men demonstrated equal sit and reach test scores as women but the women had 8 percent greater pelvic flexion (118). Men tend to possess higher musculotendinous stiffness (89, 140), which would increase the resistance to a higher ROM. One study reported 44 percent greater gastrocnemius stiffness in men (139). Lower female passive muscle stiffness may be attributed to lesser female muscle cross-sectional area and thickness (111, 123) or an intrinsically more compliant female muscle (lower viscoelastic properties) (139). Female tendons also have greater compliance than men (110) with the greater female tendon compliance associated with their oestrogen secretion (35). Women are reported to have lower collagen fibril concentrations and percent area compared to males impacting the elasticity modulus (83). Collagen is a major component of skin, fascia, cartilage, ligaments, skin, tendons and bones. It is a protein composed of a triple helix, giving it high tensile strength. Reported decreased knee joint stiffness in women was attributed to differences in hormonal concentrations (e.g., oestrogen, progesterone) affecting ligamentous properties (more compliant as a result of less collagen) and associated with a greater Q angle (40). The Q angle is defined by drawing a line from the centre of the patella to the anterior superior iliac spine and another line from the tibial tubercle through the centre of the patella. The normal female value for this angle is 13–18 degrees and 12–15 degrees for males. Greater than normal Q angles could lead to chondromalacia patella, increased incidence of knee injuries or other symptoms.

Although sex differences in hormonal concentrations have been attributed to sex differences in joint flexibility, the literature is not unanimous regarding their effects on stretching. A recent study showed that significant acute increases in passive and active knee extension following static stretching (three x 45 seconds) were not influenced by the phases of the menstrual cycle (129). However, a Japanese study found that knee extension ROM and passive torque at the onset of pain were significantly

increased during the ovulatory and luteal phases compared with the follicular phase. Passive stiffness decreased significantly during the ovulatory phase compared with the follicular phase (134). They reported that increases in muscle strength and activation (EMG) during the luteal phase was not related to flexibility, which suggested a greater implication for neural mechanisms.

Other endocrine differences occur with pregnancy. Women become more flexible during pregnancy owing to the release of various hormones such as relaxin, which allows greater extensibility of the interpubic ligament, however not all studies agree (6). Without this increased flexibility, the relatively large skull of the baby would never make it through the vaginal canal. This relaxin-induced increase in flexibility can also affect the ROM of other joints throughout the body (6).

After an acute bout of passive stretching (135 seconds), women demonstrated greater ROM increases, which were attributed to a greater stretch tolerance, as their musculotendinous stiffness did not change with the stretching (89). A study from Indonesia detected that static stretching was more effective for women whereas dynamic stretching was more favourable for men (115). Generally, in order to obtain similar ROM increases as women, men may need to stretch at a higher intensity or longer duration (89).

Ageing Differences

Ageing or senescence has been described as a “process of unfavourable progressive change” (112). Some older adults as they look in the mirror might counter that it is a depressive (psychologically and physiologically) rather than progressive change that alters their previously, younger, wrinkle free skin, decreases their strength, power and speed and increases their joint, ligament and musculotendinous stiffness. Older adults tend to be less flexible than their younger counterparts (60, 70). However, when they participate in stretch training programmes, their relative increases in flexibility are similar to young adults (60). Furthermore, trained older adults demonstrate greater degrees of flexibility than untrained older adults (60). As with other physical fitness parameters, the typically more sedentary lifestyle of senior adults exacerbates the decline in flexibility. A lack of activity can affect the synthesis, degradation and interconnections between connective tissue.

Collagen are long, fibrous structural proteins with high tensile strength and is the main component of fascia, cartilage, ligaments, tendons, bone and skin (36) (see Figure 3.8). It is an evolutionary ancient protein involved with the binding of cells of the simplest animals, such as sponges, as well as humans (11). Collagen has an extremely low compliance, similar to the tensile characteristics of copper. Intermolecular cross-links stabilize collagen, preventing the long rod-like molecules from sliding past each other forming nearly inextensible fibres (12). In combination with the proteins, elastin and soft keratin, the mix of these proteins provides not only strength (collagen) but also

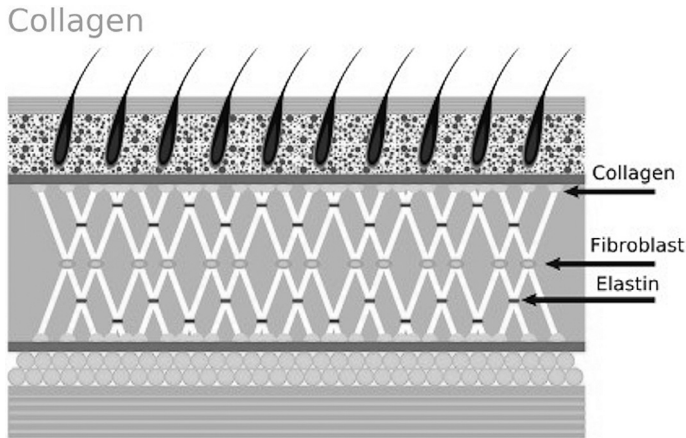


Figure 3.8 Collagen and elastin fibres within a tissue

elasticity (elastin) (5). Unlike collagen, elastin can double its length (12). With ageing, collagen increases markedly (5), which would have significant effects on ROM. The low compliance of collagen increases muscle, tendon and ligament stiffness. Thus, with increased stiffness or lower compliance, higher passive tension occurs for smaller increases in musculotendinous length. These changes would also have significant effects upon the stretch-shortening cycle. An increase in connective tissue proteins like collagen impede the muscle contraction/relaxation process, as it would have less extensible and compressible spring-like capabilities (5). A further complication is the excessive formation of intermolecular cross-links (11). Collagen and elastin cross-links in younger people promote strength and elasticity to the tissue but excessive cross-links with ageing can ensure that stiffness predominates over compliance. Another age-related change is the decreased hydration (water content) of aged tissue. Proteoglycans (i.e., chondroitin sulphate and keratin), which are present in virtually all extracellular matrices of connective tissues can hold large amounts of water, and so changes in their composition could lead to mild dehydration and some loss of function and extensibility (11). An example of this lack of extensibility with age can be seen with a simple test. Have a young person (especially a child) and a senior adult pinch the skin on the back of the hand. Then, quickly release the skin and be aware of the time it takes for the skin to return to its original shape and position. In a young child, it is practically instantaneous and would be almost impossible to measure with a stop watch or timer. With a senior adult, you can easily see and measure the slow return of the pinched skin to its original position. The older person's more dehydrated, cross linked, collagen predominant skin lacks the elasticity of the young person's skin. The same processes occur in the older person's connective tissues subcutaneously (under the skin) with the muscles, tendons, ligaments, and other tissues.

However as mentioned, this decreased ROM is attenuated in trained and more active older adults and with stretch training programmes. Coincidentally, animal studies have shown that the soleus muscles of rats do not get stiffer or as stiff with ageing and possess relatively, the same stiffness as young rat soleus muscles (5). How can this be? Rats in laboratories love to run on their little treadmills. The soleus muscle would be one of their most highly active muscles. Thus, just like active or trained humans, if the muscle activity is maintained throughout life, the degree of musculotendinous degradation and stiffness is decreased. In a typical inactive older adult, the more elastic-like elastin proteins degrade at a faster rate and are replaced by collagen. In senior aged, trained individuals, there would be less degradation, more elastin protein synthesis, less collagen replacement, and less unsuitable cross linkages. In addition, trained older adults have greater stretch tolerance (i.e., greater tolerance of passive joint moment), so they can push themselves farther while tolerating the relative discomfort (31). For older adults to remain flexible, it is recommended to participate in flexibility exercises at least two days each week for at least ten minutes each day (72). As you will read in the resistance training chapter, participating in resistance training even without stretching can enhance ROM (3). Whether older adults use traditional resistance training or functional training (multi-articular and multiplanar exercises to improve movement ability, central body strength and neuromuscular efficiency), both provided similar improvements in joint mobility over a 12-week programme (49). An American training programme called the Strong Women Programme (progressive resistance training, balance training, and flexibility exercises) showed significant sit and reach flexibility increases over their twice weekly, 12-week programme (175).

Youth Differences

It is generally considered that children have better flexibility than adults. However, more age-specific research has shown different patterns with pre-pubescence and pubescence. Younger children around the age of five years typically show a high level of flexibility which declines progressively until the age of 12 years. During puberty until pre-adulthood (12–18 years), their flexibility exhibits improvements (38, 76, 99, 105). Even with very young children when comparing 5-year-old versus 6-year-old children (64) or kindergarten to second grade (132) there has been evidence of decreases in flexibility. Not all studies provide a consistent picture. In contrast to the early studies that reported improved flexibility throughout puberty, others have reported decrements (45, 55, 68, 108, 138, 173) or no significant change (37, 128). Similar to senior adults who experience decrements in strength, power and ROM, much of the impairments can be related to their activity levels. A similar relationship seems to exist with children with the less active experiencing great ROM deficits. Young and pubescent children who partake in extensive

flexibility programmes such as gymnasts, dancers, figure skaters and others can exhibit astounding levels of flexibility, continually improving from young childhood and through puberty. Other, less active children may lose their natural flexibility. owing to the time spent sitting in school and at home (59, 133). Stretching programmes can start very early especially for extreme ROM sports like gymnastics. A flexibility programme for 5–6-year-old female gymnasts called “Game stretching”, which involves the use of balls, hoops, ribbons, maces, and other items while pretending to imitate different animals, flowers or actions (e.g., fish, dolphin, starfish, mouse, tiger, “frog”, butterfly, ballerina, cobra, snail, swing, swimmer, ostrich, fox, boa, and others) was effective in improving the ROM of various joints (51). Youth classical dancers (~nine years old) improved their passive hip flexion ROM similarly when implementing either static or PNF stretching (147).

It has been suggested that diminished flexibility during puberty as well as so-called “growth pains”, and “tightness” might be attributed to a greater growth rate of the skeletal bones compared to the growth rate of muscles and connective tissue (9, 92), (113, 130). However, a number of studies dispute this common belief citing in one study that older adolescents actually had greater flexibility (163) and a study of 600 13–14-year-old students indicated that whereas a decrease in flexibility is associated with growth, growth does not reduce flexibility (59). Furthermore, “growth pains” normally occur prior to the peak height velocity stage and thus have no connection with growth (145). Regardless of the association or lack of association with growth, one study reported that stretching reduced the duration of “growing pains” in 5–14-year-old children (18).

Genetic Differences

Claude Bouchard originally from Quebec, Canada is internationally renowned for his early genetic studies with twins. His review (34) examined the effect of genetics on flexibility. The flexibility relationship in his studies and others were quite low. Whereas two studies from the same researcher reported moderate to strong correlations between 11–15-year-old male twins (0.69) (106) and 12–17-year-old male and female twins for trunk, hip and shoulder flexibility (0.7–0.91) (106), most other twin studies found weak correlations (0.18–0.43) (42, 53, 159, 160). These results again point to the importance of the environment (nature vs. nurture); that is, the activity levels and flexibility training of individuals plays a more important role than their genetics for affecting ROM.

Limb Dominance Differences

A similar activity-related rationale may be attributed to reports of differences in ROM with dominant versus non-dominant limbs. Two tennis

studies (41, 44) reported decreased internal rotation of the shoulder but increased external rotation ROM of the dominant arm. Two baseball studies reported either no major differences between right and left sides (77) versus greater hip flexion and internal rotation of the stance leg with baseball pitchers (188). There is conflict in the literature with some studies showing insignificant or small side ROM differences with pubescent and young adult females' shoulders and lower extremities (105) and no differences in ankle dorsiflexion and plantar flexion ROM with 15–34-year-olds (142). Other studies report lateralized differences with less right wrist mobility and left hip rotation (4), and greater dominant limb humeral head retroversion angle (109). Rather than limb side differences in ROM being due to a lateralized predisposition, it seems that differences are more likely attributed to a specific, unilateral, expanded dynamic ROM (increased ROM) or higher incidence of injury to a predominant limb during repetitive tasks (decreased ROM).

Circadian (Diurnal: time of day) Differences

Restricted ROM seems to be more prevalent in the morning (17, 32, 54, 69, 150, 152, 153, 166, 194). As the individual becomes more active during the day and increases their core temperature, there will be decreases in visco-elasticity (thixotropic effects: see next section for more details), increased tissue compliance and less resistance to motion. Diurnal fluctuations in endocrine responses (i.e., epinephrine, norepinephrine, thyroid hormones, testosterone, insulin-like growth factors, growth hormone) (107) would also impact basal metabolic activity contributing to core heat flux changes during the day. As the individual becomes less active and the endocrine activity subsides later in the evening in preparation for sleeping, flexibility also diminishes (69). In order to maintain, a suitable ROM towards the evening, an individual would need to maintain or increase activity in an attempt to maintain a higher body temperature and sustain the energy facilitating hormones (i.e., epinephrine, norepinephrine, thyroid hormones, testosterone, insulin-like growth factors, growth hormone and others).

Summary

Flexibility, defined as the ROM around a joint can be altered with passive and active static, dynamic and PNF stretching as well as other techniques. These stretching techniques can provide elastic (acute or non-permanent) or plastic (semi-permanent with prolonged training) changes to joint flexibility. The majority of studies show that static and PNF stretching provide greater increases in ROM versus dynamic stretching. Yoga is also very effective for improving ROM, although yoga integrates more diverse activities than just stretching (i.e., breathing techniques, static and prolonged static contractions and meditation among other activities), which can affect overall relaxation

(enhanced parasympathetic stimulation). Normative data for joint ROM are provided in a number of texts and articles. Most measurement techniques such as the use of goniometers, fleximeters, inclinometers, photography, sit (stand) and reach tests display high reliability but may not provide similar between device values.

Females tend to have greater joint ROM than men, owing to anatomical differences, less musculotendinous unit stiffness (greater compliance), and endocrine differences. With ageing, people tend to become less flexible which may be related to increased collagen proportions and protein cross-linkages. However, much of the flexibility impairments can be attributed to greater inactivity. Youth, for their part, tend to have higher flexibility levels which may decrease during puberty compared to childhood. Once again, though, the lower levels of flexibility with puberty can be related to more inactivity. Regarding the effect of genetics on flexibility, activity again plays a more important role than DNA as studies have shown weak to strong correlations. Individuals tend to be less flexible in the morning, as core temperature is lower and the relatively long period of inactivity tends to restrict movement.

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